

**FEDERAL BOARD OF INTERMEDIATE AND SECONDARY
EDUCATION, ISLAMABAD**

Annual Examination 2026 — ANSWER KEY (Version 1)

Biology — Class IX

**Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle
| Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec
7.1-7.6)**

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Question (shortened)	Answer	Explanation
1	Father of Taxonomy	C	Carolus Linnaeus (1707-1778) introduced binomial nomenclature and is called the Father of Taxonomy. Aristotle was first to classify but not the formal father. Carl Woese proposed the three-domain system much later.
2	Three-domain system proposed by	B	Carl Woese (1990) based his three-domain system (Archaea, Bacteria, Eukarya) on rRNA nucleotide sequence analysis. Robert Whittaker proposed the five-kingdom system; Robert Hooke discovered cells.
3	Powerhouse of the cell	D	Mitochondria produce ATP through cellular respiration. They have a double membrane, cristae, and their own DNA. Ribosomes make proteins; Golgi apparatus packages; Lysosomes digest.
4	Longest phase of mitosis	C	Prophase is the longest phase. During prophase chromatin condenses into visible chromosomes, the nucleolus disappears, and the spindle apparatus begins to form.
5	Levels smallest to largest	C	Cell, tissue, organ, organ system, organism is the correct ascending order. Each level gains emergent properties not present at the level below.
6	Stable constant internal environment	C	Homeostasis is the tendency to maintain a stable, relatively constant internal environment. It is controlled through negative feedback loops involving sensors, control centres, and effectors.
7	Major bioelement constituting 99% protoplasm	B	Phosphorus is one of the six major bioelements (O, C, H, N, Ca, P) that together constitute 99% of protoplasm. Sodium, Iron, and Magnesium are minor bioelements (10 elements constituting only 1%).
8	Peptide bonds between	B	Amino acids are linked by peptide bonds to form polypeptide chains. Each amino acid has an amino group (-NH ₂) and a carboxyl group (-COOH). Bond between fatty acids is ester bond; between nucleotides is phosphodiester bond.
9	Enzymes are globular proteins that	B	Speed up biochemical reactions — enzymes are biological catalysts. They do not store hereditary information (DNA does), provide long-term energy (lipids do), or transport oxygen (haemoglobin does).
10	Competitive inhibitor occupies	B	Active site — in competitive inhibition the inhibitor is structurally similar to the substrate so it competes for and occupies the active site. Non-competitive inhibitors bind to the allosteric site.
11	ATP made up of Adenine, Ribose, and	C	Three phosphate groups — Adenine + Ribose = Adenosine; Adenosine + 1P = AMP; + 2P = ADP; + 3P = ATP. The bonds between phosphate groups are high-energy bonds shown by wavy lines.
12	Vector for Malaria	C	Female Anopheles mosquito is the vector. Ronald Ross (1897) proved this link. Plasmodium is the actual pathogen (parasite). Aedes transmits dengue; Culex transmits West Nile virus.

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

2(i) Define Biology; THREE major fields with examples 3 marks

Biology is the branch of science that deals with the study of life and living organisms (Greek: *bios* = life, *logos* = study). 1 mark

Three major fields: 1 mark each = 2 marks

- **Botany:** Study of plants (e.g., Wheat, Rose)
- **Zoology:** Study of animals (e.g., Horse, Frog)
- **Microbiology:** Study of microorganisms (e.g., Bacteria, Viruses)

OR

Six steps of the biological method 3 marks

1. Recognition of Problem | 2. Observation (collecting data) | 3. Hypothesis (testable explanation) | 4. Deduction (if...then...) | 5. Experiment (testing deduction) | 6. Results / Conclusion (accept or reject hypothesis)

0.5 mark for each correct step in sequence (6 × 0.5 = 3 marks)

2(ii) Hypothesis vs Theory vs Law 3 marks

Term	Definition	Example
Hypothesis	A tentative, testable explanation for an observation 1 mark	Anopheles mosquito may transmit Malaria
Theory	A well-tested explanation supported by much experimental evidence 1 mark	Cell theory
Law	A statement of fact that describes a natural phenomenon always observed to be true 1 mark	Law of conservation of energy

OR

Ronald Ross and Malaria 3 marks

Ronald Ross was a British physician who, in **1897**, proved that **Malaria** is caused by **Plasmodium** and is transmitted to humans through the bite of the **female Anopheles mosquito**. 1 mark He demonstrated the parasite's life cycle inside the mosquito's gut. 1 mark This discovery established the concept of a **vector** (an organism that transmits a pathogen) and earned Ross the Nobel Prize in Medicine in 1902. 1 mark

2(iii) Binomial nomenclature — definition and rules 3 marks

Binomial nomenclature is the system of giving each organism a two-part scientific name introduced by Carolus Linnaeus. 1 mark

Three rules: 2 marks

- The name consists of two parts: genus (first word, capitalized) and species (second word, lowercase).
- The entire name is written in italics when printed, or underlined when handwritten.
- The name is in Latin or Latinized form. Example: *Homo sapiens* (Human being).

OR

Three domains of Carl Woese 3 marks

Carl Woese (1990) classified life into three domains based on **rRNA nucleotide sequence** analysis. 1 mark

- **Archaea:** Prokaryotes living in extreme environments (hot springs, salty lakes).
- **Bacteria:** Common prokaryotes (e.g., *E. coli*, *Streptococcus*). 1 mark
- **Eukarya:** All eukaryotes (Protista, Fungi, Plantae, Animalia — including humans). 1 mark

2(iv) Cell membrane — functions and composition 3 marks

The cell membrane (plasma membrane) follows the **fluid mosaic model** and is composed of **60-80% proteins** and **20-40% phospholipids**. It is **semipermeable**. 1 mark

Three functions: 2 marks

- Acts as a **selective barrier** controlling what enters and leaves the cell.
- Contains **receptor proteins** for chemical signals (hormones, etc.).
- Involved in **cell recognition** and communication between cells.

OR

SER vs RER 3 marks

Feature	Rough ER (RER)	Smooth ER (SER)
Ribosomes	Present — gives rough appearance 1 mark	Absent 1 mark
Function	Protein synthesis and transport	Lipid synthesis, detoxification
Location	Near nucleus	Throughout cytoplasm 1 mark

2(v) Significance of mitosis — three headings 3 marks

- **Genetic stability:** Mitosis produces daughter cells with the same number of chromosomes as the parent cell, maintaining genetic identity across generations of cells. 1 mark
- **Growth and development:** Multicellular organisms grow from a single zygote into trillions of cells through repeated mitosis, increasing cell number in tissues and organs. 1 mark
- **Wound healing / Cell replacement:** Damaged or worn-out cells are replaced by mitosis. For example, red blood cells are replaced every 120 days and skin cells are constantly renewed. 1 mark

OR

Cytokinesis in animal vs plant cells 3 marks

Feature	Animal cells	Plant cells
Mechanism	Cleavage furrow — cell membrane pinches inward 1 mark	Cell plate (phragmoplast) forms midway between daughter nuclei 1 mark
Direction	Outside inward	Inside outward 1 mark
Reason for difference	No rigid wall	Rigid cell wall cannot be pinched

2(vi) Crossing over — location and importance 3 marks

Crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes. 1 mark It occurs during **Prophase I of Meiosis I**, at points called **chiasmata**. 1 mark Its importance is that it generates **new combinations of alleles** (genetic recombination), increasing genetic variation among offspring and providing raw material for evolution. 1 mark

OR

Benign vs malignant tumour; metastasis 3 marks

A **tumour** is a mass of cells resulting from uncontrolled cell division. 1 mark

- **Benign tumour:** Localized, does not spread to other parts; usually not life-threatening (e.g., warts). 1 mark
- **Malignant tumour (cancer):** Spreads to other tissues. The process by which cancer cells break away and invade other parts of the body is called **metastasis**. 1 mark

2(vii) Tissues forming the stomach 3 marks

The stomach contains all four types of animal tissue: 1 mark

- **Epithelial tissue:** Inner lining; secretes mucus, HCl, and pepsin enzyme. 0.5 mark
- **Connective tissue:** Supports and connects the mucosa to the muscular layer; the submucosa delivers blood supply. 0.5 mark
- **Muscle tissue (smooth):** Contracts to move food through the gut (peristalsis). 0.5 mark
- **Nervous tissue:** Controls smooth muscle contraction and secretion of digestive substances. 0.5 mark

OR

Respiratory and cardiovascular systems cooperating 3 marks

The **respiratory system** (nose, trachea, lungs) delivers air to sites where gas exchange can occur. 1 mark

The **cardiovascular system** (heart, blood vessels) picks up oxygen from the lungs, transports it to all body cells, and carries carbon dioxide back to the lungs for expulsion. 1 mark Only when both systems work together can oxygen and carbon dioxide be successfully exchanged between cells and the environment, maintaining homeostasis of blood gases. 1 mark

2(viii) Three types of plant tissue 3 marks

- **Dermal tissue:** Covers and protects the plant surface; forms the outer skin (epidermis) of leaves, stems, and roots. It prevents water loss. 1 mark
- **Ground tissue:** Serves as the site for photosynthesis (in leaves), provides support, and stores water and sugars (in stems and roots). 1 mark
- **Vascular tissue:** Transports water, minerals, and food to different parts. Includes xylem (water/minerals upward) and phloem (sugars downward). 1 mark

OR

Leaf cross-section structure 3 marks

- **Palisade mesophyll:** Two to three layers of cylindrical cells with many chloroplasts; main site of photosynthesis. 1 mark
- **Spongy mesophyll:** Loosely arranged irregular cells with large intercellular spaces; facilitates gas diffusion. 0.5 mark
- **Guard cells:** Pairs around stomata; control opening/closing for gas exchange and water vapour loss. 0.5 mark
- **Xylem vessels:** Long dead cells transporting water from root to leaf. **Phloem cells:** Carry prepared food from leaf to other parts. 1 mark

2(ix) Energy yield of lipids vs carbohydrates vs proteins; functions of lipids 3 marks

Energy per gram: **Lipids = 9.1 kcal**, Carbohydrates = 4 kcal, Proteins = 4 kcal. Lipids provide double the energy of carbohydrates because they have a higher carbon-hydrogen ratio. 1 mark

Three functions of lipids: 2 marks

- **Component of cell membranes** (phospholipids form the bilayer).
- **Energy storage** in fat cells and liver — one gram yields 9.1 kcal.
- **Steroid hormones** (e.g., testosterone, aldosterone) are lipid-derived.

OR

Classification of carbohydrates 3 marks

Group	Example	Function
Monosaccharides 1 mark	Glucose (C ₆ H ₁₂ O ₆)	Direct energy source; produced in photosynthesis
Disaccharides 1 mark	Sucrose (cane sugar)	Transport sugar in plants; common sweetener
Polysaccharides 1 mark	Starch / Glycogen	Long-term energy storage in plants / animals

2(x) Three main points of Watson and Crick DNA model 3 marks

- The DNA molecule is **double stranded**; both strands are coiled around each other like a **double helix** (like a twisted ladder). 1 mark
- Both strands are **antiparallel** (oppositely oriented); the backbone is made of sugar-phosphate and the rungs are made of nitrogen bases. 1 mark
- **Complementary base pairing**: Adenine (A) always pairs with Thymine (T) via 2 hydrogen bonds; Cytosine (C) always pairs with Guanine (G) via 3 hydrogen bonds. The diameter of the double helix is about 2 nm. 1 mark

OR**Three types of RNA and their roles** 3 marks

- **mRNA (Messenger RNA)**: Carries the message (genetic code) from DNA in the nucleus to ribosomes in the cytoplasm for protein synthesis. 1 mark
- **tRNA (Transfer RNA)**: Transfers specific amino acids to the ribosome during translation, matching codon to anticodon. 1 mark
- **rRNA (Ribosomal RNA)**: Component of ribosomes; combines with ribosomal proteins to form the ribosome — the factory of protein synthesis. 1 mark

2(xi) Enzyme inhibition — competitive vs non-competitive 3 marks

Enzyme inhibition is the phenomenon where certain chemicals at the reaction site prevent enzymes from performing their catalytic role. The chemical substance that inhibits enzyme action without being converted to products is called an **inhibitor**. 1 mark

Feature	Competitive Inhibition	Non-competitive Inhibition
Site of binding	Active site 0.5 mark	Allosteric site 0.5 mark
Similarity to substrate	Structurally similar to substrate	Does not resemble substrate
Reversibility	Reversible — overcome by increasing substrate	Not reversible by increasing substrate
Effect on active site	Blocks substrate binding	Alters shape of active site indirectly 1 mark

OR**Four factors affecting enzyme activity; temperature effect** 3 marks

Four factors: Temperature, pH, Substrate concentration, and presence of Inhibitors/Activators. 1 mark

Effect of temperature: Enzyme activity increases with temperature up to the **optimum temperature** (36–38°C for human enzymes). 1 mark Beyond this, high temperature causes the globular structure and active site to be damaged — a process called **denaturation**. At very low temperatures, enzymes are inactive due to insufficient activation energy but regain activity when temperature is restored. 1 mark

SECTION C — Long Question Answers (4 × 5 = 20 Marks)

3(i) Levels of biological organization — atomic to organism 5 marks

At each level of organization, new emergent properties appear that were not present at lower levels.

1 mark

Level	Explanation	Example
Atomic	Smallest unit of an element maintaining its properties	Carbon, Hydrogen, Oxygen 0.5 mark
Molecular	Atoms combine to form molecules with entirely different properties	Water, DNA, Glucose 0.5 mark
Organelle	Biomolecules assemble to form organelles (sub-cellular structures)	Nucleus, Ribosome 0.5 mark
Cellular	Organelles combine to form cells — smallest unit of life	Muscle cell, Neuron 0.5 mark
Tissue	Groups of similar cells performing the same function	Muscle tissue, Epithelial tissue 0.5 mark
Organ	Two or more tissues organized to carry out a specific function	Heart, Stomach, Leaf 0.5 mark
Organ system	Group of organs performing related functions	Cardiovascular system, Digestive system 0.5 mark
Organism	Several organ systems functioning together as a living individual	Human being, Rose plant 0.5 mark

OR**Advantages of classification; taxonomic hierarchy for Homo sapiens** 5 marks**Advantages of classification:** 2 marks

- Makes study of organisms easy and systematic.
- Helps understand evolutionary relationships.
- Provides a universal language for scientists worldwide.
- Facilitates identification of new species and organisms.

Taxonomic hierarchy for Human being (Homo sapiens): 3 marks

Domain : Eukarya Kingdom : Animalia Phylum : Chordata Class : Mammalia Order : Primates Family : Hominidae Genus : Homo Species : sapiens

The classification goes from most general (Domain) to most specific (Species). A species is defined as a group of organisms that can interbreed and produce fertile offspring.

3(ii) Cell organelles: Mitochondria, Chloroplast, Golgi apparatus 5 marks

(a) Mitochondria 2 marks

- **Structure:** Double-membrane organelle. Outer membrane is smooth; inner membrane is folded into projections called **cristae**. The fluid inside is called **matrix**. Contains its own DNA and ribosomes.
- **Function:** Site of cellular respiration; produces ATP (energy currency). Called the "powerhouse of the cell". One glucose molecule yields 36 ATP in aerobic respiration.

(b) Chloroplast 2 marks

- **Structure:** Double-membrane plastid containing a system of membrane sacs called **thylakoids**; stacks of thylakoids form **grana**. The fluid surrounding grana is the **stroma**.
- **Function:** Site of photosynthesis. Light reactions occur in thylakoid membranes; Calvin cycle (dark reactions) occurs in stroma. Converts light energy into chemical energy (glucose).

(c) Golgi apparatus 1 mark

- **Structure:** Stack of flattened membrane sacs called **cisternae**, discovered by Camillo Golgi.
- **Function:** Receives proteins from RER, processes/modifies them, and packages them into **vesicles** for secretion or delivery to lysosomes.

OR

Complete process of mitosis 5 marks

Mitosis produces two genetically identical daughter cells with the same chromosome number as the parent. It consists of karyokinesis (nuclear division) and cytokinesis (cytoplasm division). 0.5 mark

INTERPHASE (preparation) DNA replication in S phase; protein synthesis in G2 | PROPHASE (longest stage) Chromatin condenses into chromosomes Nucleolus disappears; spindle apparatus forms | METAPHASE Chromosomes align at the equatorial plate (metaphase plate) Spindle fibres attach to centromeres | ANAPHASE Spindle fibres pull sister chromatids to opposite poles Cell elongates | TELOPHASE Nuclear membranes reform around each set of chromosomes Nucleoli reappear; chromosomes uncoil | CYTOKINESIS Animals: cleavage furrow pinches inward Plants: cell plate (phragmoplast) forms centre outward

1 mark per stage (4 stages + cytokinesis difference = 5 marks)

3(iii) Proteins — structure, functions, sources, structural levels 5 marks

Introduction: Proteins are main structural and functional components of cells, making up about 55% of dry cell weight. Proteins are polymers of amino acids synthesized by ribosomes. 0.5 mark

Amino acids are the building blocks (monomers) of proteins. There are about 20–25 types. Each amino acid has a central alpha (α) carbon with: amino group ($-\text{NH}_2$), carboxyl group ($-\text{COOH}$), hydrogen atom ($-\text{H}$), and a variable R group that determines its properties. Amino acids are linked by **peptide bonds**. 1 mark

Four structural levels: 2 marks

- **Primary:** Linear polypeptide with a specific number, type, and sequence of amino acids.
- **Secondary:** Polypeptide folds upon itself due to extra bonds (alpha-helix or beta-sheet).
- **Tertiary:** Further folding of secondary structure with even more extra bonds to form a globular shape.
- **Quaternary:** Two or more polypeptides bonded together (e.g., haemoglobin).

Functions: Enzymes (speed up reactions), Hormones (e.g., insulin), Antibodies (immunity), Haemoglobin (O_2 transport), Structural proteins (collagen, keratin). 1 mark

Sources: Plant seeds (beans, lentils, nuts), Animal products (meat, eggs, milk, fish), and some fungi (mushrooms). 0.5 mark

OR

Central dogma: Transcription and Translation 5 marks

The flow of genetic information from DNA to RNA to protein is called the **central dogma of molecular biology**. 1 mark

DNA --[Transcription]--> mRNA --[Translation]--> Protein (Nucleus) (Nucleus/Cytoplasm) (Ribosome)

Transcription: 2 marks

- First step of gene expression occurring in the **nucleus**.
- Both strands of a gene unwind; only one strand (the coding strand) is copied.
- The nucleotide sequence of the coding DNA strand is copied as **mRNA**.
- After copying, both DNA strands re-coil into the double helix.
- mRNA then moves from nucleus to cytoplasm.

Translation: 2 marks

- Occurs at **ribosomes** in the cytoplasm.
- mRNA codons are read by ribosomes; each codon specifies one amino acid.
- **tRNA** molecules carry specific amino acids to the ribosome, matching anticodon to codon.
- Ribosome links amino acids by peptide bonds to form a growing polypeptide chain.
- All three types of RNA (mRNA, tRNA, rRNA) are involved. Together, transcription and translation are called **gene expression**.

3(iv) Mechanism of enzyme action; characteristics; regulation by inhibitors/activators 5 marks

Mechanism of enzyme action (ES/EP complex): 2 marks

Substrate + Enzyme $\rightarrow$ ES Complex $\rightarrow$ EP Complex $\rightarrow$ Product + Enzyme (Active site (Catalysis (Enzyme binds substrate) occurs) unchanged)

- The specific region on the enzyme where substrate binds is the **active site** — a charge-bearing cavity with a specific shape.
- Substrate binds to active site to form the **Enzyme-Substrate (ES) complex**.
- The enzyme catalyzes conversion of substrate to product, forming the **Enzyme-Product (EP) complex**.
- The EP complex breaks up, releasing the product. The **enzyme remains unchanged** and can be reused.

Key characteristics of enzymes: 2 marks

- **Globular proteins** (some are RNA — ribozymes)
- **Speed up reactions** millions of times without being consumed
- **Highly specific** — each enzyme catalyzes only one type of reaction
- **Required in small quantity** — unchanged after reaction, reusable
- **Lower activation energy** — enable reactions to occur at body temperature
- **May require cofactors** (activators: metal ions; coenzymes: NAD; prosthetic groups: FAD)

Regulation: 1 mark **Inhibitors** are molecules that stop enzymatic reactions (competitive: block active site; non-competitive: bind allosteric site and alter active site shape). **Activators** are molecules that increase enzyme activity. Enzyme synthesis inside cells can also be increased or decreased as needed.

OR

ATP — energy currency; ATP-ADP cycle; structure; high-energy bonds 5 marks

ATP as energy currency: 1 mark All life processes (movement, development, reproduction, active transport, thermoregulation) depend on energy. Living organisms store and use energy at the cellular level as **Adenosine Triphosphate (ATP)**. ATP is called the "energy currency of the cell" because it stores and releases energy like money — it can be used for any cellular work.

Structure of ATP: 2 marks

- **Adenine** (double-ringed nitrogen base)
- **Ribose** (five-carbon sugar) — Adenine + Ribose = Adenosine
- **Three phosphate groups (PO₄)** linked in a chain called a **triphosphate group**

Adenosine + 1P = AMP; + 2P = ADP; + 3P = ATP

High-energy bonds: The covalent bonds between phosphate groups are called **high-energy bonds**, represented by wavy lines (~). These bonds store large amounts of energy that cells use to carry out their functions. 1 mark

ATP-ADP cycle: 1 mark

Energy from catabolism/light | ADP + Phosphate $\xrightarrow{\text{[condensation]}}$ ATP (energy stored) | ATP + H₂O $\xrightarrow{\text{[hydrolysis]}}$ ADP + Pi (energy released for cellular work)

Human cells hydrolyze 100–150 moles of ATP per day. One glucose molecule in aerobic respiration generates a net of **36 ATP** molecules.